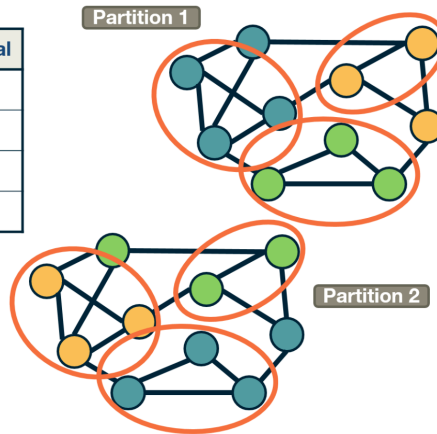


L8: Normalized Mutual Information (NMI) Metric

Confusion Matrix	C^2_1	C^2_2	C^2_3	Total
C^1_1	0	3	1	4
C^1_2	1	0	2	3
C^1_3	3	0	0	3
Total	4	3	3	10



Example: Confusion matrix with two partitions.

Suppose that we have two community partitions, P_1 and P_2 , of the same network. It may be that each partition was generated by a different community detection algorithm – or that one of the two partitions corresponds to the ground truth.

The question we now address is: how can we compare the two partitions?

To make such comparisons in a statistical

manner, we need the joint distribution $p(C^1_i, C^2_j)$ of the two partitions, i.e., the probability that a node belongs to the i 'th community C^1_i of the first partition and the j 'th community C^2_j of the second partition.

The marginal distribution $p(C^1_i)$ of the first partition represents the probability that a node belongs to community C^1_i of the first partition. Similarly for the marginal $p(C^2_j)$ of the second partition.

The “**confusion matrix**” (see the visualization) can be used to calculate the previous joint distribution – as well as the two marginal distributions (*last row and last column*).

A common way to compare two statistical distributions in Information Theory is through the Mutual Information – this metric quantifies how much information we get for one of two random variables if we know the value of the other. If the two random variables are statistically independent, we do not get any information. If the two random variables are identical, we get the maximum information.

When the two random variables represent the two community partitions of the same network, the definition of Mutual Information is written as:

$$\sum_{i,j} p(C^1_i, C^2_j) \log_2 \frac{p(C^1_i, C^2_j)}{p(C^1_i)p(C^2_j)}$$

To get a metric that varies between 0 (statistically independent partitions) and 1 (identical partitions), we need to normalize the mutual information with the average entropy of the two partitions,

$$\frac{1}{2} [H(p(C^1)) + H(p(C^2))]$$

Recall that $H(p(X))$ is the entropy of the distribution $p(X)$ of random variable X , defined as:

$$H(X) = - \sum_x p(x) \log_2 p(x)$$

So, to summarize, the Normalized Mutual Information metric (NMI) is defined as:

$$\text{NMI} = I_n = \frac{\sum_{i,j} p(C^1_i, C^2_j) \log_2 \frac{p(C^1_i, C^2_j)}{p(C^1_i)p(C^2_j)}}{\frac{1}{2} [H(p(C^1)) + H(p(C^2))]}$$

Food for Thought

Show that the minimum value of the mutual information metric results when the two partitions are statistically independent, and that the maximum value results when they are identical. Also, what is that maximum value?.